Deep Learning 秋 2026
Homework 3

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3.1

$$
\begin{aligned}
D_f &= \int q\left(-\left(\frac{p}{q}+1\right)\log\frac{\frac{p}{q}+1}{2}+\frac{p}{q}\log\frac{p}{q}\right) \\
&= \int q\left(-\frac{p}{q}\log\frac{\frac{\frac{p}{q}+1}{2}}{\frac{p}{q}}-\log\frac{1+\frac{p}{q}}{2}\right) \\
&= \int q\left(-\frac{p}{q}\log\frac{p+q}{2p}-\log\frac{p+q}{2q}\right) \\
&= \int p\log\frac{p}{\frac{p+q}{2}}+q\log\frac{q}{\frac{p+q}{2}} \\
&= \mathrm{JS}(p,q)
\end{aligned}
$$

3.2

$$
f\left(\frac{p_X}{p_G}\right)=\sup\left\{\frac{p_X}{p_G}t-f^*(t)\right\}=\frac{p_X}{p_G}D^*-f^*D^*
$$

thus, D^* follows above equation.

3.3

$$
f^*(t)=\sup\{xt-f(x)\}=x^*t-f(x^*)
$$

$$
\frac{\partial(xt-f(x))}{\partial x}=t-(\log x-\log(x+1))\Rightarrow x^*=\frac{e^t}{1-e^t}
$$

thus,

$$
\begin{aligned}
f^*(t) &= x^*t-f(x^*) \\
&= -\log(1-e^t)
\end{aligned}
$$

$$
f\left(\frac{p_X}{p_G}\right)=\frac{p_X}{p_G}\log\frac{p_X}{p_X+p_G}+\log\frac{p_G}{p_G+p_X}
$$

thus,$D^*=\log\frac{p_X}{p_X+p_G}$ set f in $V(G,D^*)$,we have $V(G,D^*)=\mathrm{JS}(p,q)$

3.4

$$
C^\top=\begin{pmatrix}\omega_1 & 0\\ \omega_2 & 0\\ \omega_3 & \omega_1\\ 0 & \omega_2\\ 0 & \omega_3\end{pmatrix}
$$

3.5

$$\begin{aligned}\mathrm{KL}(p(x,y),p_c(x,y)) &= \mathbb{E}_p\left(\log\frac{p(x,y)}{p_c(x,y)}\right)\\ &= \mathbb{E}_p\left(\log\frac{p(x)p(y|x)}{p_c(x)p_c(y|x)}\right)\\ &= \mathbb{E}_p\log\frac{p(y|x)}{p_c(y|x)}\\ &= \text{Constant} - \mathbb{E}_p\log p_c(y|x)\end{aligned}$$

3.6

关于 C 最小化R_p 等价与最小化$\mathrm{KL}(p_g,p_c)$.因此最小化该项等价与

$$\min_{p_g}\min_{p_c} R_p = \min_{p_g} R_p|_{p_c=p_g}$$

因此最终$p=p_c=p_g$